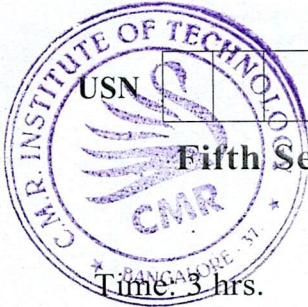




**Fifth Semester B.E./B.Tech. Degree Examination, June/July 2025**  
**Electromagnetic Waves**

21EC54

Max. Marks: 100

**Note :** Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What do you mean by scalar and vector fields? Show the difference between two. (06 Marks)  
 b. Given three points in Cartesian coordinate system as A(3, -2, 1), B(-3, -3, 5), C(2, 6, -4).  
 Find : i) The vector from A to C  
 ii) The unit vector from B to A  
 iii) The distance from B to C  
 iv) The vector from A to the midpoint of the straight line joining B to C. (08 Marks)  
 c. State Coulomb's law of force between any two point charges and also in vector form. (06 Marks)

OR

- 2 a. A charge  $Q_2 = 12 \text{ inc}$  is located in free space at  $P_2 (-0.03, 0.01, 0.04)\text{m}$ . Find the force on  $Q_2$  due to  $Q_1$  where  $Q_1 = 110\mu\text{C}$  at  $P_1(0.03, 0.08, -0.02)\text{m}$ . (06 Marks)  
 b. A volume charge density is expressed as  $\rho_v = 10z^2 x \sin \pi y$ . Find the total charge inside the volume  $(-1 \leq x \leq 2), (0 \leq y \leq 1), (3 \leq z \leq 3.6)$ . (06 Marks)  
 c. Derive the expression for electric field intensity due to infinite line charge. (08 Marks)

Module-2

- 3 a. State and prove the Gauss's law. (06 Marks)  
 b. Consider a coaxial cable with inner radius 'a' and outer radius 'b'. Derive the expression for flux density ( $\bar{D}$ ) for the region  $a < r < b$  using Gauss's law. (08 Marks)  
 c. The flux density  $\bar{D} = r/3 \bar{a}_r \text{ nc/m}^2$  is in the free space :  
 i) Find  $\bar{E}$  at  $r = 0.2\text{m}$   
 ii) Find the electric flux leaving the sphere of  $r = 0.2\text{m}$ .  
 iii) Find the total charge within the sphere of  $r = 0.3\text{m}$ . (06 Marks)

OR

- 4 a. Derive Maxwell first equation as applied to the electro statics, using Gauss's law. State the divergence theorem using Maxwell's first equation. (06 Marks)  
 b. Evaluate the both sides of divergence theorem for the field  $\bar{D} = 2xy \bar{a}_x + x^2 \bar{a}_y \text{ c/m}^2$  and rectangular parallel piped formed by the planes  $x = 0$  and  $x = 1$ ,  $y = 0$  and  $y = 2$  and  $z = 0$  and  $z = 3$ . (08 Marks)  
 c. Derive the expression for the work done in moving a point charge in an electric field. (06 Marks)

Module-3

- 5 a. Determine whether or not the following potential fields satisfy the Laplace's equation :  
 i)  $V = x^2 - y^2 + z^2$  ii)  $V = r \cos \phi + z$  iii)  $V = r \cos \theta + \phi$ . (06 Marks)  
 b. Using the Laplace's equation, derive an expression for capacitance per unit length of a coaxial cable using the following boundary conditions.  $V = V_0$  at  $r = a$  and  $V = 0$  at  $r = b$ ,  $b > a$ . (08 Marks)  
 c. State and explain Biot – Savart law applicable to magnetic field. (06 Marks)

OR

- 6 a. Derive the expression for a curl, applying Ampere's circuital law to an incremental surface element. (08 Marks)  
 b. State and prove the Stoke's theorem. (06 Marks)  
 c. What is scalar magnetic potential? Explain Laplace equations for scalar magnetic potential. (06 Marks)

Module-4

- 7 a. Define and explain the terms magnetic flux and magnetic flux density. Obtain the magnetic flux using magnetic flux density in coaxial cable. (08 Marks)  
 b. In certain region, the magnetic flux density in a magnetic material with  $\chi_m = 6$  is given and  $\bar{B} = 0.005y^2 \bar{a}_x \text{ T}$ . At  $y = 0.4\text{m}$ , find the magnitude of : i)  $\bar{J}$  ii)  $\bar{J}_b$  iii)  $\bar{J}_T$ . (06 Marks)  
 c. Discuss the boundary conditions for magnetic field based on the normal component of the  $\bar{B}$  and  $\bar{H}$ . (06 Marks)

OR

- 8 a. Derive an expression for the magnetic force between differential current elements. (06 Marks)  
 b. A conductor of length 2.5m in  $z = 0$  and  $x = 0$  carries a current of 12A in  $-\bar{a}_y$  direction. Calculate the uniform flux in the region, if the force on the conductor is  $12 \times 10^{-2} \text{ N}$  in the direction specified by  $\left[ \frac{-\bar{a}_x + \bar{a}_z}{\sqrt{2}} \right]$ . (08 Marks)  
 c. State and explain Faraday's law of electromagnetic induction in integral and point form. (06 Marks)

Module-5

- 9 a. Write the Maxwell's equations in the integral form and explain the physical significance. (06 Marks)  
 b. Two parallel conducting plates of area  $0.05\text{m}^2$  are separated by 2mm of lossy, dielectric for which  $\epsilon_r = 8.3$  and  $\sigma = 8 \times 10^{-4} \text{ S/m}$ . given an applied voltage  $V = 10 \sin 10^7 t \text{ V}$ . Find total r.m.s current. (08 Marks)  
 c. Do the fields  $\bar{E} = E_m \sin x \sin t \bar{a}_y$  and  $\bar{H} = \frac{E_m}{\mu_0} \cos x \cos t \bar{a}_z$  satisfy the Maxwell's equations. (06 Marks)

OR

- 10 a. Write short notes on Retarded potential. (06 Marks)  
 b. Given  $\bar{E} = E_0 z^2 e^{-t} \bar{a}_x$  in free space, determine if there exist a magnetic field such that both Faraday's law and Ampere's circuital law are satisfied simultaneously. (08 Marks)  
 c. Discuss the propagation of uniform plane wave in good conductor. (06 Marks)

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